

VK-RCD 实验结果汇总（中文实验部分）

作者信息待补充

2026-05-22

1 实验设置

本文仅汇总实验部分，不包含完整引言、相关工作与方法推导。实验基于 MM-Fi/X-Fi 风格多模态协议：HPE 使用 VK、Depth、LiDAR、mmWave、WiFi-CSI，报告 MPJPE 与 PA-MPJPE，单位为 mm，越低越好；HAR 使用 VK、Depth、LiDAR、mmWave，报告 Accuracy 与 Macro-F1，越高越好。表中 I/VK 表示官方 X-Fi 的 I 为 RGB，本文对应位置使用 VK，不将二者视为完全同源输入。Teacher 是全模态特权知识源；缺失模态部署结论主要依据 Student-VK 与 Student-NV。

HPE 的相对提升定义为 $(\text{Official} - \text{Ours})/\text{Official}$ ；HAR 的相对提升定义为 $(\text{Ours} - \text{Official})/\text{Official}$ 。蓝色表示优于官方或优于参考项，红色表示退化。官方 X-Fi 数值来自本地 legacy CSV 与公开代码仓库 <https://github.com/NTUMARS/X-Fi>。

2 HPE 定量结果

2.1 Random Split 主结果

表 1: HPE random split 全模态/非视觉主结果与官方 X-Fi 对比。Student-NV 不含 VK，因此与官方 D+L+R+W 对比。

方法	组合	MPJPE	X-Fi MPJPE	Imp	PA-MPJPE	X-Fi PA	Imp
Baseline	I/VK+D+L+R+W	49.14	83.70	+41.30%	33.48	47.60	+29.66%
Teacher	I/VK+D+L+R+W	48.45	83.70	+42.11%	33.06	47.60	+30.54%
Student-VK	I/VK+D+L+R+W	50.17	83.70	+40.06%	33.48	47.60	+29.66%
Student-NV	D+L+R+W	51.03	97.60	+47.72%	35.69	47.40	+24.69%

2.2 Cross-Scene 迁移

表 2: HPE cross-scene 与 random split 对比。 Δ 为 cross-scene 减 random，越小越好。

方法	组合	Random MPJPE	Cross MPJPE	Δ	Random PA	Cross PA	Δ
Teacher	I/VK+D+L+R+W	48.45	76.06	+27.61	33.06	37.89	+4.83
Student-VK	I/VK+D+L+R+W	50.17	71.70	+21.53	33.48	47.74	+14.26
Student-NV	D+L+R+W	51.03	78.96	+27.93	35.69	46.38	+10.69

2.3 任意模态组合汇总

表 3: HPE Student-VK 31 种组合按可用模态数量汇总。Best/Worst 使用 MPJPE 排序。

可用数	组合数	MPJPE	X-Fi	Imp	PA	X-Fi PA	Imp	Best	Worst
1	5	124.76	143.16	+12.85%	68.01	77.40	+12.13%	D / 53.88	W / 222.14
2	10	78.10	104.97	+25.60%	46.85	59.72	+21.56%	I/VK+D / 51.65	L+W / 134.16
3	10	60.78	92.58	+34.35%	38.08	52.32	+27.21%	I/VK+D+R / 50.50	L+R+W / 94.17
4	5	53.55	87.94	+39.11%	34.89	49.58	+29.62%	I/VK+D+L+R / 50.25	I/VK+L+R+W / 64.25
5	1	50.17	83.70	+40.06%	33.48	47.60	+29.66%	I/VK+D+L+R+W / 50.17	I/VK+D+L+R+W / 50.17
平均	31	75.18	103.70	+27.51%	45.07	58.16	+22.50%	I/VK+D+L+R+W / 50.17	W / 222.14

3 HAR 定量结果

3.1 Random、Cross-Scene 与 Cross-Subject 主结果

表 4: HAR 主结果与官方 X-Fi 对比。为避免 final_eval metadata 混淆, 本表统一取 all-combinations 评估中的完整组合行。

协议	方法	组合	Acc	Macro-F1	X-Fi Acc	Imp
random	Teacher	I/VK+D+L+R	95.57	95.56	72.20	+32.37%
random	Student-VK	I/VK+D+L+R	96.56	96.55	72.20	+33.74%
random	Student-NV	D+L+R	96.12	96.13	80.50	+19.41%
cross-scene	Teacher	I/VK+D+L+R	94.80	94.83	72.20	+31.30%
cross-scene	Student-VK	I/VK+D+L+R	95.25	95.25	72.20	+31.92%
cross-scene	Student-NV	D+L+R	95.41	95.40	80.50	+18.52%
cross-subject	Teacher	I/VK+D+L+R	95.52	95.48	72.20	+32.30%
cross-subject	Student-VK	I/VK+D+L+R	96.12	96.10	72.20	+33.13%
cross-subject	Student-NV	D+L+R	95.70	95.67	80.50	+18.88%

3.2 HAR 组合鲁棒性汇总

表 5: HAR Student-VK 15 组合与 Student-NV 7 个非视觉组合的协议级汇总。Best/Worst 使用 Acc 排序。

协议	方法	组合数	Avg Acc	X-Fi Avg	Imp	Avg F1	Best	Worst
random	Student-VK	15	85.15	62.46	+36.32%	85.15	I/VK+D+L+R / 96.56	L / 40.80
cross-scene	Student-VK	15	80.61	62.46	+29.06%	80.35	I/VK+D+R / 95.32	L / 19.98
cross-subject	Student-VK	15	85.30	62.46	+36.56%	85.46	I/VK+D+L+R / 96.12	L / 38.46
random	Student-NV	7	80.62	69.59	+15.86%	80.48	D+R / 96.18	L / 26.12
cross-scene	Student-NV	7	76.70	69.59	+10.23%	76.41	D+R / 95.45	L / 8.94
cross-subject	Student-NV	7	81.26	69.59	+16.77%	80.82	D+R / 95.76	L / 25.30

4 消融、鲁棒性与复杂度

4.1 HPE 消融

表 6: HPE 全模态消融。 Δ 相对 Student-VK 完整目标, MPJPE/PA 越低越好。

设置	MPJPE	Δ	PA-MPJPE	Δ	说明
Student-VK 完整目标	50.17	+0.00	33.48	+0.00	随机缺失 + 蒸馏 + 可靠性融合
NoKD	48.83	-1.35	32.51	-0.97	移除 Teacher 蒸馏
Output KD	47.62	-2.55	32.15	-1.33	仅输出蒸馏
Output+Token KD	48.76	-1.42	33.24	-0.24	输出与 token 蒸馏
Output+Bone KD	49.34	-0.84	33.38	-0.10	输出与骨架结构蒸馏
Full Structural KD	49.20	-0.97	34.63	+1.15	完整结构蒸馏变体
Uniform Fusion	52.28	+2.11	37.22	+3.74	等权融合

4.2 HAR 消融

表 7: HAR random split 消融。Full Δ 相对 Student-VK, Acc 越高越好。

设置	Full Acc	Δ	Full F1	Avg-15 Acc	Avg-15 F1
Student-VK	96.54	+0.00	96.53	85.15	85.15
NoKD	95.97	-0.57	95.95	82.86	82.56
Uniform	95.40	-1.13	95.40	67.46	66.44

4.3 HPE VK 噪声与关节缺失鲁棒性

表 8: Student-VK 在 VK 噪声与关节随机丢弃下的鲁棒性。 Δ 相对无噪声/无丢弃。

噪声标准差	关节丢弃率	MPJPE	Δ	PA-MPJPE	Δ
0.0	0.0	50.18	+0.00	33.48	+0.00
0.0	0.1	52.64	+2.47	34.57	+1.09
0.0	0.3	54.10	+3.93	35.19	+1.71
0.0	0.5	54.75	+4.57	35.50	+2.02
5.0	0.0	50.88	+0.70	33.92	+0.45
5.0	0.1	52.88	+2.70	34.71	+1.23
5.0	0.3	54.12	+3.94	35.19	+1.71
5.0	0.5	54.76	+4.59	35.50	+2.02
10.0	0.0	52.08	+1.90	34.57	+1.09
10.0	0.1	53.26	+3.09	34.93	+1.45
10.0	0.3	54.12	+3.94	35.20	+1.72
10.0	0.5	54.74	+4.57	35.52	+2.04
20.0	0.0	53.63	+3.45	35.20	+1.72
20.0	0.1	53.85	+3.67	35.15	+1.67
20.0	0.3	54.21	+4.03	35.25	+1.77
20.0	0.5	54.78	+4.61	35.54	+2.06

4.4 模型复杂度

表 9: HPE 已同步模型复杂度。HAR 当前 final/eval CSV 含参数量，但未同步有效 FPS/显存。

模型	组合	Params	FPS	Peak Memory
Baseline	I/VK+D+L+R+W	5.74M	52.04	4.60 GB
Teacher	I/VK+D+L+R+W	8.90M	63.24	4.61 GB
Student-VK	I/VK+D+L+R+W	8.90M	62.01	4.61 GB
Student-NV	D+L+R+W	8.90M	63.37	4.61 GB

5 实验结论与不足

1. 在 HPE random split 中，Student-VK 在全模态下达到约 50.17 mm MPJPE，相比官方全模态 83.70 mm 有明显下降；Student-NV 在不使用 VK 的 D+L+R+W 下也优于官方非视觉组合，说明非 RGB 部署路线具备可行性。
2. Teacher 的全模态精度较强，但其缺失模态组合结果只用于诊断；面向任意模态缺失部署时，应优先报告随机缺失训练后的 Student-VK 与 Student-NV。
3. HPE cross-scene 明显高于 random split，说明环境迁移仍是 HPE 的主要挑战；目前 HPE cross-subject 结果未完整同步，不能写成已完成结论。
4. HAR 在 random、cross-scene、cross-subject 的 Student-VK/Student-NV 结果整体较强，尤其 Student-NV 说明无视觉输入也能保持较高动作识别性能。
5. 消融结果表明蒸馏策略与可靠性融合需要分任务讨论：HAR 中 NoKD/Uniform 仍强，但低于 Student-VK；HPE 中部分蒸馏变体可能优于默认 Student-VK，需要后续用统一随机种子和重复实验确认显著性。

A HPE Student-VK Random 31 组合全表

组合	MPJPE	X-Fi	Imp	PA-MPJPE	X-Fi PA	Imp
D	53.88	101.80	+47.08%	35.61	48.40	+26.44%
L	140.14	167.10	+16.13%	93.52	103.20	+9.38%
R	114.98	127.40	+9.75%	58.13	69.80	+16.71%
I/VK	92.66	93.90	+1.32%	42.60	60.30	+29.35%
W	222.14	225.60	+1.53%	110.19	105.30	-4.64%
D+L	53.00	102.50	+48.29%	35.35	48.40	+26.96%
D+R	52.32	98.00	+46.62%	34.64	47.30	+26.77%
D+W	53.54	101.80	+47.40%	35.52	48.10	+26.15%
L+R	97.52	109.80	+11.18%	55.08	63.40	+13.12%
L+W	134.16	159.50	+15.89%	93.18	102.70	+9.27%
R+W	108.54	117.20	+7.39%	57.27	62.70	+8.66%
I/VK+D	51.65	86.10	+40.01%	34.29	48.10	+28.70%
I/VK+L	73.18	93.00	+21.31%	41.92	59.70	+29.79%

组合	MPJPE	X-Fi	Imp	PA-MPJPE	X-Fi PA	Imp
I/VK+R	68.93	88.80	+22.38%	38.90	57.30	+32.12%
I/VK+W	88.11	93.00	+5.26%	42.30	59.50	+28.90%
D+L+R	51.92	96.00	+45.92%	34.50	47.30	+27.06%
D+L+W	52.85	102.00	+48.18%	35.38	48.10	+26.44%
D+R+W	52.13	97.00	+46.26%	34.62	47.10	+26.50%
L+R+W	94.17	107.40	+12.32%	55.05	63.10	+12.75%
I/VK+D+L	51.10	84.80	+39.74%	34.16	48.20	+29.14%
I/VK+D+R	50.50	83.40	+39.45%	33.52	47.30	+29.12%
I/VK+D+W	51.63	85.30	+39.47%	34.29	48.10	+28.70%
I/VK+L+R	66.08	88.40	+25.25%	38.77	57.20	+32.23%
I/VK+L+W	70.73	93.00	+23.95%	41.76	59.70	+30.05%
I/VK+R+W	66.69	88.50	+24.64%	38.80	57.10	+32.06%
D+L+R+W	51.78	97.60	+46.95%	34.54	47.40	+27.13%
I/VK+D+L+R	50.25	83.50	+39.82%	33.45	47.60	+29.73%
I/VK+D+L+W	51.05	86.00	+40.64%	34.18	48.20	+29.08%
I/VK+D+R+W	50.41	84.00	+39.98%	33.54	47.60	+29.55%
I/VK+L+R+W	64.25	88.60	+27.49%	38.75	57.10	+32.13%
I/VK+D+L+R+W	50.17	83.70	+40.06%	33.48	47.60	+29.66%

B HPE Student-VK Cross-Scene 31 组合全表

组合	MPJPE	X-Fi	Imp	PA-MPJPE	X-Fi PA	Imp
D	71.91	101.80	+29.36%	47.74	48.40	+1.36%
L	500.56	167.10	-199.56%	162.44	103.20	-57.40%
R	155.81	127.40	-22.30%	75.70	69.80	-8.45%
I/VK	144.98	93.90	-54.39%	61.46	60.30	-1.92%
W	355.26	225.60	-57.47%	112.24	105.30	-6.59%
D+L	79.79	102.50	+22.15%	50.35	48.40	-4.02%
D+R	68.84	98.00	+29.75%	46.03	47.30	+2.69%
D+W	72.45	101.80	+28.83%	47.34	48.10	+1.59%
L+R	283.93	109.80	-158.59%	115.53	63.40	-82.23%
L+W	484.16	159.50	-203.55%	160.80	102.70	-56.57%
R+W	153.26	117.20	-30.77%	75.32	62.70	-20.13%
I/VK+D	67.16	86.10	+22.00%	45.92	48.10	+4.53%
I/VK+L	151.56	93.00	-62.97%	80.49	59.70	-34.82%
I/VK+R	115.13	88.80	-29.65%	56.42	57.30	+1.54%
I/VK+W	127.91	93.00	-37.54%	60.55	59.50	-1.76%
D+L+R	77.47	96.00	+19.30%	48.84	47.30	-3.26%
D+L+W	79.94	102.00	+21.63%	50.22	48.10	-4.41%
D+R+W	69.70	97.00	+28.14%	46.20	47.10	+1.90%
L+R+W	278.03	107.40	-158.87%	115.21	63.10	-82.59%
I/VK+D+L	73.32	84.80	+13.54%	48.09	48.20	+0.22%
I/VK+D+R	66.24	83.40	+20.57%	45.47	47.30	+3.87%
I/VK+D+W	67.86	85.30	+20.45%	46.05	48.10	+4.27%
I/VK+L+R	126.79	88.40	-43.42%	72.75	57.20	-27.18%
I/VK+L+W	140.48	93.00	-51.05%	80.43	59.70	-34.73%
I/VK+R+W	109.38	88.50	-23.59%	56.30	57.10	+1.41%
D+L+R+W	77.38	97.60	+20.72%	48.90	47.40	-3.17%
I/VK+D+L+R	71.62	83.50	+14.23%	47.64	47.60	-0.08%
I/VK+D+L+W	73.50	86.00	+14.54%	48.18	48.20	+0.05%
I/VK+D+R+W	66.77	84.00	+20.51%	45.69	47.60	+4.00%
I/VK+L+R+W	119.89	88.60	-35.32%	72.42	57.10	-26.83%
I/VK+D+L+R+W	71.71	83.70	+14.33%	47.76	47.60	-0.34%

C HPE Student-NV 非视觉组合全表

协议	组合	MPJPE	X-Fi	Imp	PA-MPJPE	X-Fi PA	Imp
random	D	52.34	101.80	+48.59%	36.44	48.40	+24.72%
random	L	123.10	167.10	+26.33%	84.16	103.20	+18.45%
random	R	109.44	127.40	+14.10%	57.49	69.80	+17.63%
random	W	212.40	225.60	+5.85%	106.87	105.30	-1.50%
random	D+L	52.03	102.50	+49.24%	36.39	48.40	+24.81%
random	D+R	51.24	98.00	+47.71%	35.86	47.30	+24.18%
random	D+W	52.46	101.80	+48.47%	36.36	48.10	+24.41%
random	L+R	91.13	109.80	+17.01%	54.08	63.40	+14.71%
random	L+W	123.65	159.50	+22.48%	84.02	102.70	+18.19%
random	R+W	106.06	117.20	+9.51%	57.44	62.70	+8.39%
random	D+L+R	51.11	96.00	+46.76%	35.75	47.30	+24.42%
random	D+L+W	52.02	102.00	+49.00%	36.37	48.10	+24.39%
random	D+R+W	51.22	97.00	+47.20%	35.77	47.10	+24.06%
random	L+R+W	90.93	107.40	+15.34%	54.14	63.10	+14.21%
random	D+L+R+W	51.03	97.60	+47.72%	35.69	47.40	+24.70%
cross-scene	D	73.59	101.80	+27.71%	41.34	48.40	+14.59%
cross-scene	L	294.85	167.10	-76.45%	137.42	103.20	-33.16%
cross-scene	R	138.70	127.40	-8.87%	67.71	69.80	+2.99%
cross-scene	W	372.60	225.60	-65.16%	118.38	105.30	-12.42%
cross-scene	D+L	82.29	102.50	+19.72%	48.46	48.40	-0.12%
cross-scene	D+R	71.38	98.00	+27.16%	41.03	47.30	+13.25%
cross-scene	D+W	74.08	101.80	+27.23%	41.38	48.10	+13.97%
cross-scene	L+R	218.15	109.80	-98.68%	111.45	63.40	-75.80%
cross-scene	L+W	297.00	159.50	-86.21%	138.56	102.70	-34.92%
cross-scene	R+W	155.05	117.20	-32.29%	68.75	62.70	-9.66%
cross-scene	D+L+R	78.60	96.00	+18.13%	46.25	47.30	+2.23%
cross-scene	D+L+W	81.72	102.00	+19.88%	48.25	48.10	-0.31%
cross-scene	D+R+W	72.42	97.00	+25.34%	41.06	47.10	+12.81%
cross-scene	L+R+W	218.96	107.40	-103.87%	112.27	63.10	-77.93%
cross-scene	D+L+R+W	78.96	97.60	+19.10%	46.38	47.40	+2.14%

D HAR Student-VK 15 组合全表

协议/模型	组合	Acc	X-Fi	Acc	Imp	Macro-F1
random Student-VK	D	88.02	48.10		+83.00%	87.86
random Student-VK	L	40.80	52.70		-22.59%	40.10
random Student-VK	R	85.88	85.70		+0.21%	86.06
random Student-VK	I/VK	65.31	26.50		+146.47%	65.70
random Student-VK	D+L	88.52	51.60		+71.54%	88.39
random Student-VK	D+R	96.45	79.80		+20.86%	96.44
random Student-VK	L+R	89.51	88.70		+0.91%	89.57
random Student-VK	I/VK+D	89.51	45.30		+97.59%	89.36
random Student-VK	I/VK+L	68.54	35.20		+94.73%	69.05
random Student-VK	I/VK+R	92.75	73.40		+26.36%	92.82
random Student-VK	D+L+R	96.44	80.50		+19.80%	96.43
random Student-VK	I/VK+D+L	89.62	48.70		+84.01%	89.49
random Student-VK	I/VK+D+R	96.56	70.70		+36.57%	96.55
random Student-VK	I/VK+L+R	92.73	77.80		+19.19%	92.81
random Student-VK	I/VK+D+L+R	96.56	72.20		+33.74%	96.55
cross-scene Student-VK	D	87.41	48.10		+81.73%	87.30
cross-scene Student-VK	L	19.98	52.70		-62.09%	17.51
cross-scene Student-VK	R	80.30	85.70		-6.30%	80.17
cross-scene Student-VK	I/VK	57.99	26.50		+118.82%	57.53
cross-scene Student-VK	D+L	87.65	51.60		+69.86%	87.49
cross-scene Student-VK	D+R	94.94	79.80		+18.97%	94.93
cross-scene Student-VK	L+R	81.15	88.70		-8.51%	81.42
cross-scene Student-VK	I/VK+D	88.84	45.30		+96.11%	88.79
cross-scene Student-VK	I/VK+L	59.39	35.20		+68.72%	58.81
cross-scene Student-VK	I/VK+R	88.76	73.40		+20.93%	88.65
cross-scene Student-VK	D+L+R	94.87	80.50		+17.86%	94.87
cross-scene Student-VK	I/VK+D+L	88.95	48.70		+82.64%	88.88
cross-scene Student-VK	I/VK+D+R	95.32	70.70		+34.82%	95.32
cross-scene Student-VK	I/VK+L+R	88.33	77.80		+13.53%	88.34
cross-scene Student-VK	I/VK+D+L+R	95.25	72.20		+31.92%	95.25
cross-subject Student-VK	D	90.05	48.10		+87.21%	89.96
cross-subject Student-VK	L	38.46	52.70		-27.03%	36.69
cross-subject Student-VK	R	85.21	85.70		-0.58%	85.45
cross-subject Student-VK	I/VK	65.22	26.50		+146.13%	67.65
cross-subject Student-VK	D+L	90.28	51.60		+74.97%	90.17
cross-subject Student-VK	D+R	95.71	79.80		+19.93%	95.68
cross-subject Student-VK	L+R	88.10	88.70		-0.67%	88.14
cross-subject Student-VK	I/VK+D	91.58	45.30		+102.17%	91.51
cross-subject Student-VK	I/VK+L	68.95	35.20		+95.88%	70.90
cross-subject Student-VK	I/VK+R	93.12	73.40		+26.87%	93.12
cross-subject Student-VK	D+L+R	95.74	80.50		+18.94%	95.71
cross-subject Student-VK	I/VK+D+L	91.72	48.70		+88.33%	91.64
cross-subject Student-VK	I/VK+D+R	96.10	70.70		+35.92%	96.07
cross-subject Student-VK	I/VK+L+R	93.06	77.80		+19.62%	93.06
cross-subject Student-VK	I/VK+D+L+R	96.12	72.20		+33.13%	96.10

E HAR Student-NV 非视觉 7 组合全表

协议/模型	组合	Acc	X-Fi	Acc	Imp	Macro-F1
random Student-NV	D	88.57	48.10		+84.14%	88.47
random Student-NV	L	26.12	52.70		-50.43%	25.01
random Student-NV	R	83.37	85.70		-2.72%	83.57
random Student-NV	D+L	88.68	51.60		+71.85%	88.56
random Student-NV	D+R	96.18	79.80		+20.53%	96.19
random Student-NV	L+R	85.30	88.70		-3.83%	85.38
random Student-NV	D+L+R	96.12	80.50		+19.41%	96.13
cross-scene Student-NV	D	88.82	48.10		+84.65%	88.79
cross-scene Student-NV	L	8.94	52.70		-83.03%	6.39
cross-scene Student-NV	R	79.70	85.70		-7.01%	79.99
cross-scene Student-NV	D+L	88.92	51.60		+72.32%	88.87
cross-scene Student-NV	D+R	95.45	79.80		+19.62%	95.45
cross-scene Student-NV	L+R	79.68	88.70		-10.16%	79.99
cross-scene Student-NV	D+L+R	95.41	80.50		+18.52%	95.40
cross-subject Student-NV	D	90.30	48.10		+87.73%	90.21
cross-subject Student-NV	L	25.30	52.70		-51.99%	22.23
cross-subject Student-NV	R	84.57	85.70		-1.31%	84.75
cross-subject Student-NV	D+L	90.43	51.60		+75.26%	90.34
cross-subject Student-NV	D+R	95.76	79.80		+20.00%	95.74
cross-subject Student-NV	L+R	86.74	88.70		-2.21%	86.76
cross-subject Student-NV	D+L+R	95.70	80.50		+18.88%	95.67

F HPE Teacher/Baseline/消融诊断组合表

模型/设置	组合	MPJPE	X-Fi	Imp	PA-MPJPE	X-Fi PA	Imp
Teacher random	D	234.48	101.80	-130.33%	84.60	48.40	-74.79%
Teacher random	L	1615.86	167.10	-867.00%	387.35	103.20	-275.34%
Teacher random	R	4196.64	127.40	-3194.06%	298.72	69.80	-327.97%
Teacher random	I/VK	487.99	93.90	-419.69%	139.35	60.30	-131.10%
Teacher random	W	4173.33	225.60	-1749.88%	378.22	105.30	-259.18%
Teacher random	D+L	285.88	102.50	-178.91%	87.48	48.40	-80.74%
Teacher random	D+R	178.87	98.00	-82.52%	65.56	47.30	-38.60%
Teacher random	D+W	197.61	101.80	-94.11%	74.07	48.10	-53.99%
Teacher random	L+R	2158.39	109.80	-1865.75%	387.74	63.40	-511.58%
Teacher random	L+W	4007.14	159.50	-2412.31%	384.12	102.70	-274.02%

模型/设置	组合	MPJPE	X-Fi	Imp	PA-MPJPE	X-Fi PA	Imp
Teacher random	R+W	4675.91	117.20	-3889.68%	273.15	62.70	-335.65%
Teacher random	I/VK+D	369.12	86.10	-328.71%	97.31	48.10	-102.31%
Teacher random	I/VK+L	1072.73	93.00	-1053.47%	213.13	59.70	-257.00%
Teacher random	I/VK+R	814.71	88.80	-817.47%	174.05	57.30	-203.76%
Teacher random	I/VK+W	2491.83	93.00	-2579.39%	335.20	59.50	-463.36%
Teacher random	D+L+R	310.47	96.00	-223.41%	75.92	47.30	-60.50%
Teacher random	D+L+W	177.40	102.00	-73.93%	66.98	48.10	-39.25%
Teacher random	D+R+W	121.88	97.00	-25.65%	54.83	47.10	-16.40%
Teacher random	L+R+W	4360.36	107.40	-3959.92%	322.27	63.10	-410.73%
Teacher random	I/VK+D+L	634.71	84.80	-648.48%	112.28	48.20	-132.94%
Teacher random	I/VK+D+R	465.16	83.40	-457.74%	86.45	47.30	-82.77%
Teacher random	I/VK+D+W	100.71	85.30	-18.06%	51.47	48.10	-7.00%
Teacher random	I/VK+L+R	1034.39	88.40	-1070.12%	237.91	57.20	-315.93%
Teacher random	I/VK+L+W	4736.11	93.00	-4992.59%	271.23	59.70	-354.32%
Teacher random	I/VK+R+W	4769.71	88.50	-5289.51%	255.98	57.10	-348.30%
Teacher random	D+L+R+W	112.99	97.60	-15.77%	50.72	47.40	-7.01%
Teacher random	I/VK+D+L+R	593.40	83.50	-610.66%	93.25	47.60	-95.90%
Teacher random	I/VK+D+L+W	67.89	86.00	+21.05%	40.53	48.20	+15.91%
Teacher random	I/VK+D+R+W	71.27	84.00	+15.16%	38.91	47.60	+18.25%
Teacher random	I/VK+L+R+W	4876.09	88.60	-5403.49%	239.18	57.10	-318.88%
Teacher random	I/VK+D+L+R+W	48.45	83.70	+42.12%	33.06	47.60	+30.54%
Baseline random	D	172.79	101.80	-69.74%	101.84	48.40	-110.41%
Baseline random	L	1367.60	167.10	-718.43%	284.00	103.20	-175.19%
Baseline random	R	420.81	127.40	-230.30%	135.17	69.80	-93.66%
Baseline random	I/VK	256.34	93.90	-172.99%	106.01	60.30	-75.80%
Baseline random	W	2151.81	225.60	-853.82%	349.26	105.30	-231.68%
Baseline random	D+L	1413.97	102.50	-1279.48%	256.89	48.40	-430.77%
Baseline random	D+R	166.92	98.00	-70.33%	61.81	47.30	-30.67%
Baseline random	D+W	2417.11	101.80	-2274.37%	361.68	48.10	-651.92%
Baseline random	L+R	510.63	109.80	-365.05%	157.61	63.40	-148.59%
Baseline random	L+W	2157.85	159.50	-1252.88%	354.90	102.70	-245.57%
Baseline random	R+W	1063.85	117.20	-807.72%	205.61	62.70	-227.92%
Baseline random	I/VK+D	77.50	86.10	+9.98%	51.37	48.10	-6.79%
Baseline random	I/VK+L	267.25	93.00	-187.36%	101.89	59.70	-70.67%
Baseline random	I/VK+R	324.61	88.80	-265.55%	99.91	57.30	-74.36%
Baseline random	I/VK+W	249.86	93.00	-168.67%	103.47	59.50	-73.91%
Baseline random	D+L+R	179.54	96.00	-87.02%	63.00	47.30	-33.19%
Baseline random	D+L+W	2233.06	102.00	-2089.27%	347.07	48.10	-621.57%
Baseline random	D+R+W	344.01	97.00	-254.65%	64.45	47.10	-36.83%
Baseline random	L+R+W	1266.99	107.40	-1079.70%	230.97	63.10	-266.04%
Baseline random	I/VK+D+L	64.51	84.80	+23.93%	42.50	48.20	+11.82%
Baseline random	I/VK+D+R	83.67	83.40	-0.32%	43.78	47.30	+7.45%
Baseline random	I/VK+D+W	76.88	85.30	+9.87%	42.12	48.10	+12.43%
Baseline random	I/VK+L+R	300.45	88.40	-239.88%	97.86	57.20	-71.09%
Baseline random	I/VK+L+W	258.28	93.00	-177.72%	100.49	59.70	-68.33%
Baseline random	I/VK+R+W	281.54	88.50	-218.13%	98.89	57.10	-73.19%
Baseline random	D+L+R+W	502.74	97.60	-415.10%	84.98	47.40	-79.28%
Baseline random	I/VK+D+L+R	62.11	83.50	+25.62%	34.75	47.60	+26.99%
Baseline random	I/VK+D+L+W	68.70	86.00	+20.12%	40.71	48.20	+15.55%
Baseline random	I/VK+D+R+W	52.64	84.00	+37.34%	34.00	47.60	+28.57%
Baseline random	I/VK+L+R+W	289.24	88.60	-226.46%	97.69	57.10	-71.09%
Baseline random	I/VK+D+L+R+W	49.14	83.70	+41.29%	33.48	47.60	+29.66%
NoKD random	D	52.46	101.80	+48.47%	34.60	48.40	+28.51%
NoKD random	L	138.35	167.10	+17.20%	94.31	103.20	+8.62%
NoKD random	R	111.03	127.40	+12.85%	56.90	69.80	+18.48%
NoKD random	I/VK	78.34	93.90	+16.57%	41.68	60.30	+30.87%
NoKD random	W	224.81	225.60	+0.35%	109.53	105.30	-4.02%
NoKD random	D+L	52.00	102.50	+49.27%	34.47	48.40	+28.78%
NoKD random	D+R	50.74	98.00	+48.22%	33.49	47.30	+29.20%
NoKD random	D+W	52.54	101.80	+48.39%	34.49	48.10	+28.30%
NoKD random	L+R	92.30	109.80	+15.94%	53.60	63.40	+15.45%
NoKD random	L+W	134.68	159.50	+15.56%	93.87	102.70	+8.59%
NoKD random	R+W	105.27	117.20	+10.18%	55.88	62.70	+10.88%
NoKD random	I/VK+D	50.52	86.10	+41.32%	33.36	48.10	+30.64%
NoKD random	I/VK+L	68.22	93.00	+26.64%	41.28	59.70	+30.85%
NoKD random	I/VK+R	66.33	88.80	+25.30%	38.27	57.30	+33.22%
NoKD random	I/VK+W	76.12	93.00	+18.15%	41.75	59.50	+29.83%
NoKD random	D+L+R	50.48	96.00	+47.42%	33.43	47.30	+29.31%
NoKD random	D+L+W	52.06	102.00	+48.96%	34.45	48.10	+28.38%
NoKD random	D+R+W	50.73	97.00	+47.71%	33.46	47.10	+28.95%
NoKD random	L+R+W	90.81	107.40	+15.45%	53.65	63.10	+14.98%
NoKD random	I/VK+D+L	50.05	84.80	+40.97%	33.30	48.20	+30.90%
NoKD random	I/VK+D+R	49.01	83.40	+41.23%	32.52	47.30	+31.25%
NoKD random	I/VK+D+W	50.50	85.30	+40.80%	33.33	48.10	+30.71%
NoKD random	I/VK+L+R	63.75	88.40	+27.89%	38.05	57.20	+33.48%
NoKD random	I/VK+L+W	67.61	93.00	+27.30%	41.30	59.70	+30.82%
NoKD random	I/VK+R+W	64.58	88.50	+27.03%	38.30	57.10	+32.92%
NoKD random	D+L+R+W	50.49	97.60	+48.27%	33.44	47.40	+29.45%
NoKD random	I/VK+D+L+R	48.83	83.50	+41.52%	32.50	47.60	+31.73%
NoKD random	I/VK+D+L+W	50.08	86.00	+41.77%	33.30	48.20	+30.91%
NoKD random	I/VK+D+R+W	48.99	84.00	+41.68%	32.52	47.60	+31.69%
NoKD random	I/VK+L+R+W	62.65	88.60	+29.28%	38.13	57.10	+33.23%
NoKD random	I/VK+D+L+R+W	48.83	83.70	+41.66%	32.51	47.60	+31.70%
Uniform random	D	55.72	101.80	+45.26%	38.62	48.40	+20.21%
Uniform random	L	147.07	167.10	+11.99%	100.53	103.20	+2.59%
Uniform random	R	114.29	127.40	+10.29%	62.21	69.80	+10.87%
Uniform random	I/VK	123.62	93.90	-31.65%	47.89	60.30	+20.58%
Uniform random	W	231.27	225.60	-2.51%	112.01	105.30	-6.37%
Uniform random	D+L	54.97	102.50	+46.37%	38.40	48.40	+20.66%
Uniform random	D+R	54.19	98.00	+44.70%	38.04	47.30	+19.59%
Uniform random	D+W	55.33	101.80	+45.65%	38.21	48.10	+20.57%
Uniform random	L+R	100.71	109.80	+8.28%	61.29	63.40	+3.33%
Uniform random	L+W	147.28	159.50	+7.66%	100.73	102.70	+1.92%
Uniform random	R+W	113.30	117.20	+3.33%	62.09	62.70	+0.97%
Uniform random	I/VK+D	53.62	86.10	+37.72%	37.69	48.10	+21.64%
Uniform random	I/VK+L	89.60	93.00	+3.66%	47.66	59.70	+20.17%
Uniform random	I/VK+R	82.24	88.80	+7.39%	44.22	57.30	+22.83%
Uniform random	I/VK+W	124.69	93.00	-34.07%	47.44	59.50	+20.27%
Uniform random	D+L+R	53.93	96.00	+43.82%	37.89	47.30	+19.89%
Uniform random	D+L+W	54.70	102.00	+46.37%	38.22	48.10	+20.54%
Uniform random	D+R+W	53.93	97.00	+44.40%	37.81	47.10	+19.72%
Uniform random	L+R+W	100.78	107.40	+6.16%	61.37	63.10	+2.75%
Uniform random	I/VK+D+L	53.31	84.80	+37.14%	37.69	48.20	+21.81%
Uniform random	I/VK+D+R	52.43	83.40	+37.13%	37.33	47.30	+21.09%

模型/设置	组合	MPJPE	X-Fi	Imp	PA-MPJPE	X-Fi PA	Imp
Uniform random	I/VK+D+W	53.46	85.30	+37.33%	37.59	48.10	+21.85%
Uniform random	I/VK+L+R	77.10	88.40	+12.78%	44.23	57.20	+22.68%
Uniform random	I/VK+L+W	91.49	93.00	+1.62%	47.46	59.70	+20.50%
Uniform random	I/VK+R+W	84.38	88.50	+4.66%	43.97	57.10	+23.00%
Uniform random	D+L+R+W	53.74	97.60	+44.94%	37.78	47.40	+20.29%
Uniform random	I/VK+D+L+R	52.37	83.50	+37.28%	37.28	47.60	+21.67%
Uniform random	I/VK+D+L+W	53.18	86.00	+38.17%	37.60	48.20	+21.98%
Uniform random	I/VK+D+R+W	52.33	84.00	+37.70%	37.22	47.60	+21.80%
Uniform random	I/VK+L+R+W	78.57	88.60	+11.32%	44.13	57.10	+22.72%
Uniform random	I/VK+D+L+R+W	52.28	83.70	+37.54%	37.22	47.60	+21.80%

G HAR Teacher/消融诊断组合表

模型/设置	组合	Acc	X-Fi Acc	Imp	Macro-F1
random Teacher	D	83.71	48.10	+74.03%	82.80
random Teacher	L	3.36	52.70	-93.63%	1.17
random Teacher	R	66.72	85.70	-22.15%	64.78
random Teacher	I/VK	3.40	26.50	-87.15%	0.45
random Teacher	D+L	83.37	51.60	+61.56%	82.54
random Teacher	D+R	95.76	79.80	+19.99%	95.74
random Teacher	L+R	66.69	88.70	-24.81%	63.96
random Teacher	I/VK+D	83.50	45.30	+84.33%	82.60
random Teacher	I/VK+L	3.91	35.20	-88.88%	1.31
random Teacher	I/VK+R	66.46	73.40	-9.46%	63.67
random Teacher	D+L+R	95.64	80.50	+18.80%	95.63
random Teacher	I/VK+D+L	83.66	48.70	+71.79%	82.91
random Teacher	I/VK+D+R	95.66	70.70	+35.30%	95.65
random Teacher	I/VK+L+R	66.39	77.80	-14.66%	63.36
random Teacher	I/VK+D+L+R	95.57	72.20	+32.37%	95.56
cross-scene Teacher	D	84.86	48.10	+76.43%	84.47
cross-scene Teacher	L	3.96	52.70	-92.49%	1.50
cross-scene Teacher	R	61.11	85.70	-28.69%	59.73
cross-scene Teacher	I/VK	4.44	26.50	-83.25%	0.79
cross-scene Teacher	D+L	84.33	51.60	+63.44%	83.84
cross-scene Teacher	D+R	95.07	79.80	+19.14%	95.10
cross-scene Teacher	L+R	60.49	88.70	-31.81%	58.23
cross-scene Teacher	I/VK+D	85.21	45.30	+88.10%	84.97
cross-scene Teacher	I/VK+L	4.01	35.20	-88.61%	1.49
cross-scene Teacher	I/VK+R	60.86	73.40	-17.09%	59.44
cross-scene Teacher	D+L+R	94.83	80.50	+17.81%	94.86
cross-scene Teacher	I/VK+D+L	84.86	48.70	+74.26%	84.57
cross-scene Teacher	I/VK+D+R	94.96	70.70	+34.32%	94.99
cross-scene Teacher	I/VK+L+R	61.15	77.80	-21.40%	59.02
cross-scene Teacher	I/VK+D+L+R	94.80	72.20	+31.30%	94.83
cross-subject Teacher	D	81.74	48.10	+69.94%	80.25
cross-subject Teacher	L	3.89	52.70	-92.63%	2.75
cross-subject Teacher	R	72.35	85.70	-15.57%	71.76
cross-subject Teacher	I/VK	5.15	26.50	-80.56%	2.00
cross-subject Teacher	D+L	82.45	51.60	+59.79%	81.23
cross-subject Teacher	D+R	95.74	79.80	+19.98%	95.71
cross-subject Teacher	L+R	69.75	88.70	-21.37%	68.99
cross-subject Teacher	I/VK+D	82.87	45.30	+82.94%	81.39
cross-subject Teacher	I/VK+L	4.11	35.20	-88.33%	1.75
cross-subject Teacher	I/VK+R	69.56	73.40	-5.23%	68.95
cross-subject Teacher	D+L+R	95.51	80.50	+18.65%	95.47
cross-subject Teacher	I/VK+D+L	82.85	48.70	+70.12%	81.62
cross-subject Teacher	I/VK+D+R	95.72	70.70	+35.39%	95.69
cross-subject Teacher	I/VK+L+R	69.26	77.80	-10.97%	68.44
cross-subject Teacher	I/VK+D+L+R	95.52	72.20	+32.30%	95.48
random NoKD	D	86.85	48.10	+80.55%	86.78
random NoKD	L	33.14	52.70	-37.11%	31.11
random NoKD	R	83.13	85.70	-3.00%	83.31
random NoKD	I/VK	60.94	26.50	+129.97%	59.31
random NoKD	D+L	87.24	51.60	+69.07%	87.25
random NoKD	D+R	95.57	79.80	+19.76%	95.56
random NoKD	L+R	85.92	88.70	-3.13%	86.09
random NoKD	I/VK+D	88.09	45.30	+94.45%	87.94
random NoKD	I/VK+L	63.63	35.20	+80.75%	62.65
random NoKD	I/VK+R	91.05	73.40	+24.05%	91.12
random NoKD	D+L+R	95.50	80.50	+18.64%	95.50
random NoKD	I/VK+D+L	88.62	48.70	+81.98%	88.53
random NoKD	I/VK+D+R	95.96	70.70	+35.73%	95.95
random NoKD	I/VK+L+R	91.23	77.80	+17.27%	91.29
random NoKD	I/VK+D+L+R	95.99	72.20	+32.96%	95.98
random Uniform	D	85.59	48.10	+77.93%	85.30
random Uniform	L	2.65	52.70	-94.97%	1.55
random Uniform	R	69.91	85.70	-18.42%	68.71
random Uniform	I/VK	5.52	26.50	-79.16%	1.32
random Uniform	D+L	86.16	51.60	+66.98%	85.91
random Uniform	D+R	95.54	79.80	+19.73%	95.54
random Uniform	L+R	67.80	88.70	-23.56%	66.56
random Uniform	I/VK+D	86.48	45.30	+90.90%	86.24
random Uniform	I/VK+L	4.58	35.20	-86.98%	1.76
random Uniform	I/VK+R	67.75	73.40	-7.69%	66.21
random Uniform	D+L+R	95.42	80.50	+18.54%	95.42
random Uniform	I/VK+D+L	86.43	48.70	+77.48%	86.19
random Uniform	I/VK+D+R	95.50	70.70	+35.07%	95.49
random Uniform	I/VK+L+R	67.09	77.80	-13.76%	65.08
random Uniform	I/VK+D+L+R	95.39	72.20	+32.12%	95.39

H 同步结果索引

任务	CSV	行数	本文使用方式
HPE	ablation_no_distill_all_combinations.csv	31	主文摘要或同步索引
HPE	ablation_uniform_fusion_all_combinations.csv	31	主文摘要或同步索引
HPE	all_combinations.csv	31	主文摘要或同步索引
HPE	baseline_full_all_combinations.csv	31	主文摘要或同步索引
HPE	baseline_full_model_complexity.csv	1	主文摘要或同步索引
HPE	baseline_missing_modality_summary.csv	5	主文摘要或同步索引
HPE	student_nv_cross_scene_nonvisual_combinations.csv	15	附录全表 + 主文跨场景
HPE	student_nv_missing_modality_summary.csv	4	主文摘要或同步索引
HPE	student_nv_model_complexity.csv	1	主文摘要或同步索引
HPE	student_nv_nonvisual_combinations.csv	15	重复 random 非视觉表，索引记录
HPE	student_nv_random_nonvisual_combinations.csv	15	附录全表
HPE	student_vk_cross_scene_all_combinations.csv	31	附录全表 + 主文跨场景
HPE	student_vk_missing_modality_summary.csv	5	主文摘要或同步索引
HPE	student_vk_model_complexity.csv	1	主文摘要或同步索引
HPE	student_vk_noise_robustness.csv	16	主文鲁棒性表
HPE	student_vk_random_all_combinations.csv	31	附录全表 + 主文组合汇总
HPE	teacher_all_combinations.csv	31	主文摘要或同步索引
HPE	teacher_cross_scene_all_combinations.csv	31	Teacher 跨场景诊断
HPE	teacher_full_model_complexity.csv	1	主文摘要或同步索引
HPE	teacher_missing_modality_summary.csv	5	主文摘要或同步索引
HPE	teacher_random_all_combinations.csv	31	Teacher 诊断，不作为部署结论
HAR	ablation_no_distill_all_combinations.csv	15	主文摘要或同步索引
HAR	ablation_uniform_fusion_all_combinations.csv	15	主文摘要或同步索引
HAR	all_combinations.csv	15	主文摘要或同步索引
HAR	student_nv_cross_scene_nonvisual_combinations.csv	7	附录全表 + 主文组合汇总
HAR	student_nv_cross_subject_nonvisual_combinations.csv	7	附录全表 + 主文组合汇总
HAR	student_nv_nonvisual_combinations.csv	7	主文摘要或同步索引
HAR	student_nv_random_nonvisual_combinations.csv	7	附录全表 + 主文组合汇总
HAR	student_vk_all_combinations.csv	15	主文摘要或同步索引
HAR	student_vk_cross_scene_all_combinations.csv	15	附录全表 + 主文组合汇总
HAR	student_vk_cross_subject_all_combinations.csv	15	附录全表 + 主文组合汇总
HAR	student_vk_random_all_combinations.csv	15	附录全表 + 主文组合汇总
HAR	teacher_cross_scene_all_combinations.csv	15	主文摘要或同步索引
HAR	teacher_cross_subject_all_combinations.csv	15	主文摘要或同步索引
HAR	teacher_random_all_combinations.csv	15	主文摘要或同步索引

I 缺失与 Metadata 审计

任务	类型	项目	状态
HPE	eval	teacher_cross_subject_all_combinations.csv	未同步或未完成
HPE	eval	student_vk_cross_subject_all_combinations.csv	未同步或未完成
HPE	eval	student_nv_cross_subject_nonvisual_combinations.csv	未同步或未完成
HPE	eval	baseline_cross_scene_all_combinations.csv	未同步或未完成
HPE	eval	baseline_cross_subject_all_combinations.csv	未同步或未完成
HPE	eval	ablation_reliability_attention_all_combinations.csv	未同步或未完成
HPE	eval	ablation_reliability_uncertainty_all_combinations.csv	未同步或未完成
HPE	eval	ablation_encoder_mlp_vk_all_combinations.csv	未同步或未完成
HPE	eval	ablation_encoder_skeleton_prompt_all_combinations.csv	未同步或未完成
HAR	metadata	student_vk_missing_cross_scene/final_eval.csv	目录为 cross-scene，但 CSV split=cross_subject_split
HAR	metadata	teacher_full_cross_subject/final_eval.csv	文件名为 Teacher，但 method=VK-RCD-HAR-Student-VK-Missing
HAR	model	student_vk_missing_cross_subject/best.pth	缺少 best.pth，但 final/eval CSV 已同步